

Exercise of Supervised Learning: Regularization Part 1

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Exercise 1: L0 Regularization

Consider the regression learning setting, i.e., $\mathcal{Y} = \mathbb{R}$, and the feature space $\mathcal{X} = \mathbb{R}^p$. Let the hypothesis space be the linear models:

$$\mathcal{H} = \{f(\mathbf{x}) = \boldsymbol{\theta}^T \mathbf{x} \mid \boldsymbol{\theta} \in \mathbb{R}^p\}.$$

Suppose your loss function of interest is the L2 loss $L(y, f(\mathbf{x})) = \frac{1}{2}(y - f(\mathbf{x}))^2$. Consider the L_0 -regularized empirical risk of a model $f(\mathbf{x} \mid \boldsymbol{\theta})$:

$$\mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \lambda \|\boldsymbol{\theta}\|_0 = \frac{1}{2} \sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^T \mathbf{x}^{(i)})^2 + \lambda \sum_{i=1}^p \mathbf{1}_{|\theta_i| \neq 0}.$$

Assume that $\mathbf{X}^T \mathbf{X} = \mathbf{I}$, which holds if $\mathbf{X}$ has orthonormal columns. Show that the minimizer $\hat{\boldsymbol{\theta}}_{L0} = (\hat{\theta}_{L0,1}, \dots, \hat{\theta}_{L0,p})^T$ is given by

$$\hat{\theta}_{L0,i} = \hat{\theta}_i \mathbf{1}_{|\hat{\theta}_i| > \sqrt{2\lambda}}, \quad i = 1, \dots, p,$$

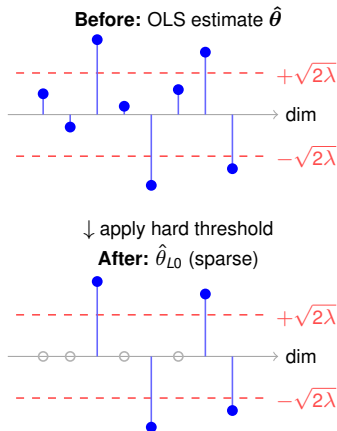
where $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \dots, \hat{\theta}_p)^T = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ is the minimizer of the unregularized empirical risk. For this purpose, using the following steps:

Roadmap: what we want to show & how

Target identity:

$$\hat{\theta}_{L0,i} = \hat{\theta}_i \cdot \mathbf{I}_{|\hat{\theta}_i| > \sqrt{2\lambda}}, \quad i = 1, \dots, p.$$

Coordinate-wise *hard thresholding* of the OLS solution $\hat{\theta}$.



Roadmap of the proof:

- (i) *Algebra.* Use $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ to rewrite $\mathcal{R}_{\text{reg}}(\boldsymbol{\theta})$ as a per-coordinate sum in $\theta_i, \hat{\theta}_i$.
- (ii) *Separability.* Splits into p independent 1D problems via $\sum_{i=1}^p g_i(\theta_i)$.
- (iii) *Strong-signal regime* $|\hat{\theta}_i| > \sqrt{2\lambda}$: determine $\arg \min_{\theta_i} g_i$.
- (iv) *Compact case form.* Encode (iii) via the indicator $\mathbf{I}_{|\hat{\theta}_i| > \sqrt{2\lambda}}$.
- (v) *Weak-signal regime* $|\hat{\theta}_i| \leq \sqrt{2\lambda}$: determine $\arg \min_{\theta_i} g_i$; assemble the full $\hat{\theta}_{L0}$.

Why the threshold is $\sqrt{2\lambda}$: it emerges in (iii)/(v) from balancing the quadratic gain $\frac{1}{2}\hat{\theta}_i^2$ against the penalty λ .

Exercise 1 (i)

(i) Derive that

$$\arg \min_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^p -\hat{\theta}_i \theta_i + \frac{\theta_i^2}{2} + \lambda \mathbf{1}_{|\theta_i| \neq 0}.$$

Note that θ_i is from the minimizer $\hat{\boldsymbol{\theta}}$ of the unregularized empirical risk :

$$\hat{\boldsymbol{\theta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{X}^T \mathbf{y}$$

because we assume that $\mathbf{X}^T \mathbf{X} = \mathbf{I}$.

Solution to Exercise 1 (i)

$$\arg \min_{\boldsymbol{\theta}} \mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \frac{1}{2} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \lambda \sum_{i=1}^p I_{|\theta_i| \neq 0}$$

Solution to Exercise 1 (i)

$$\begin{aligned}\arg \min_{\boldsymbol{\theta}} \mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \lambda \sum_{i=1}^p I_{|\theta_i| \neq 0} \\ &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2} \mathbf{y}^T \mathbf{y} - \mathbf{y}^T \mathbf{X}\boldsymbol{\theta} + \frac{1}{2} \boldsymbol{\theta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\theta} + \lambda \sum_{i=1}^p I_{|\theta_i| \neq 0}\end{aligned}$$

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Solution to Exercise 1 (i)

$$\begin{aligned}\arg \min_{\boldsymbol{\theta}} \mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) &= \arg \min_{\boldsymbol{\theta}} \frac{1}{2} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \lambda \sum_{i=1}^p I_{|\theta_i| \neq 0} \\&= \arg \min_{\boldsymbol{\theta}} \frac{1}{2} \mathbf{y}^T \mathbf{y} - \mathbf{y}^T \mathbf{X}\boldsymbol{\theta} + \frac{1}{2} \boldsymbol{\theta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\theta} + \lambda \sum_{i=1}^p I_{|\theta_i| \neq 0} \\&= \arg \min_{\boldsymbol{\theta}} - \underbrace{\mathbf{y}^T \mathbf{X} \boldsymbol{\theta}}_{\hat{\boldsymbol{\theta}}^T} + \frac{1}{2} \boldsymbol{\theta}^T \underbrace{\mathbf{X}^T \mathbf{X}}_{=\mathbf{I}} \boldsymbol{\theta} + \lambda \sum_{i=1}^p I_{|\theta_i| \neq 0} \\&= \arg \min_{\boldsymbol{\theta}} -\hat{\boldsymbol{\theta}}^T \boldsymbol{\theta} + \frac{1}{2} \boldsymbol{\theta}^T \boldsymbol{\theta} + \lambda \sum_{i=1}^p I_{|\theta_i| \neq 0}\end{aligned}$$

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Roadmap: where are we?

✓ (i) *Algebra*. Use $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ to rewrite $\mathcal{R}_{\text{reg}}(\boldsymbol{\theta})$ as a per-coordinate sum in θ_i , $\hat{\theta}_i$.

► (ii) **Separability**. Splits into p independent 1D problems via $\sum_{i=1}^p g_i(\theta_i)$.

(iii) *Strong-signal regime* $|\hat{\theta}_i| > \sqrt{2\lambda}$: determine $\arg \min_{\theta_i} g_i$.

(iv) *Compact case form*. Encode (iii) via the indicator $\mathbf{I}_{|\hat{\theta}_i| > \sqrt{2\lambda}}$.

(v) *Weak-signal regime* $|\hat{\theta}_i| \leq \sqrt{2\lambda}$: determine $\arg \min_{\theta_i} g_i$; assemble the full $\hat{\boldsymbol{\theta}}_{L0}$.

Exercise 1 (ii)

(ii) Note that the minimization problem on the right-hand side of (i) can be written as $\sum_{i=1}^p g_i(\theta)$, where

$$g_i(\theta) = -\hat{\theta}_i\theta + \frac{\theta^2}{2} + \lambda \mathbf{1}_{|\theta| \neq 0}.$$

What is the advantage of this representation if we seek to find the θ with entries $\theta_1, \dots, \theta_p$ minimizing $\mathcal{R}_{\text{reg}}(\theta)$?

Recall from (i):

$$\arg \min_{\theta} \mathcal{R}_{\text{emp}}(\theta) = \arg \min_{\theta} \sum_{i=1}^p \left(-\hat{\theta}_i\theta_i + \frac{\theta_i^2}{2} + \lambda \mathbf{1}_{|\theta_i| \neq 0} \right).$$

Solution to Exercise 1 (ii)

Advantage: minimize each g_i **separately**.

Why?

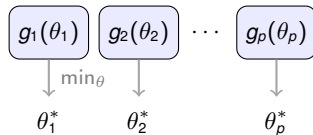
- ▶ Each g_i involves *only* θ_i (not $\theta_j, j \neq i$).
- ▶ Coordinates *decoupled*: moving θ_i doesn't touch any g_j .

Key identity — min of sum = sum of mins:

$$\min_{\theta_1, \dots, \theta_p} \sum_{i=1}^p g_i(\theta_i) = \sum_{i=1}^p \min_{\theta_i} g_i(\theta_i).$$

Payoff: p -D problem $\rightarrow p$ 1-D problems;
solve $\min_{\theta} g_i(\theta)$ once, parametric in $\hat{\theta}_i$.

Decoupled subproblems



Each g_i optimised in parallel,
independently of the others.

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- (iii) **Strong-signal regime** $|\hat{\theta}_i| > \sqrt{2\lambda}$: **determine** $\arg \min_{\theta_i} g_i$.
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Exercise 1 (iii)

(iii) Consider the first case that $|\hat{\theta}_i| > \sqrt{2\lambda}$ and infer that for the minimizer θ_i^* of g_i it must hold that $\theta_i^* = \hat{\theta}_i$.

Hint: Show that $g_i(\theta_i) < 0 = g_i(0)$ and argue that the minimizer must have the same sign as $\hat{\theta}_i$.

(Personally I find this hint is not so useful.)

In symbols:

$$\underbrace{|\hat{\theta}_i| > \sqrt{2\lambda}}_{\text{strong signal}} \implies \underbrace{\theta_i^* = \hat{\theta}_i}_{\text{regularized} = \text{unregularized}}$$

Solution to Exercise 1 (iii)

$$\text{Objective: } g_i(\theta_i) = -\hat{\theta}_i \theta_i + \frac{\theta_i^2}{2} + \lambda I_{|\theta_i| \neq 0}.$$

Split $g_i(\theta_i)$ at the jump $\theta_i = 0$.

- **Case 1:** $\theta_i = 0$
 $g_i(0) = 0$.
- **Case 2:** $\theta_i \neq 0$, so g_i is a smooth quadratic

$$g_i(\theta_i) = \frac{\theta_i^2}{2} - \hat{\theta}_i \theta_i + \lambda.$$

Minimizer $\theta_i^* = \hat{\theta}_i$, with min value

$$g_i(\hat{\theta}_i) = -\frac{\hat{\theta}_i^2}{2} + \lambda.$$

Why compare? The penalty $\lambda I_{|\theta_i| \neq 0}$ makes g_i jump at 0 — each case yields its own candidate minimum, so the global min is whichever is smaller.

Compare using $|\hat{\theta}_i|^2 > 2\lambda$:

$$\underbrace{-\frac{\hat{\theta}_i^2}{2} + \lambda}_{\text{Case 2 min}} < \underbrace{0}_{= g_i(0)}$$

Conclusion:

$$\theta_i^* = \hat{\theta}_i \text{ when } |\hat{\theta}_i| > \sqrt{2\lambda}.$$

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$$\underbrace{-\frac{\hat{\theta}_i^2}{2} + \lambda}_{\text{Case 2 min}} < \underbrace{0}_{= g_i(0)}$$

Conclusion: $\theta_i^* = \hat{\theta}_i$ when $|\hat{\theta}_i| > \sqrt{2\lambda}$.

Roadmap: where are we?

- ✓ (i) *Algebra*. Use $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ to rewrite $\mathcal{R}_{\text{reg}}(\boldsymbol{\theta})$ as a per-coordinate sum in $\theta_i, \hat{\theta}_i$.
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- (v) *Weak-signal regime* $|\hat{\theta}_i| \leq \sqrt{2\lambda}$: determine $\arg \min_{\theta_i} g_i$; assemble the full $\hat{\boldsymbol{\theta}}_{L0}$.

Exercise 1 (iv)

(iv) Derive that $\theta_i^* = \hat{\theta}_i \mathbb{I}_{|\hat{\theta}_i| > \sqrt{2\lambda}}$, by using (iii).

Setting: **still considering the case** $|\hat{\theta}_i| > \sqrt{2\lambda}$.

Solution: In the solution of (iii) we have proven that

$$\theta_i^* = \hat{\theta}_i \quad \text{if} \quad |\hat{\theta}_i| > \sqrt{2\lambda},$$

The optimal θ_i can be written as

$$\theta_i^* = \hat{\theta}_i \cdot 1 = \hat{\theta}_i \mathbb{I}_{|\hat{\theta}_i| > \sqrt{2\lambda}}.$$

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(iv) Derive that $\theta_i^* = \hat{\theta}_i \mathbf{1}_{|\hat{\theta}_i| > \sqrt{2\lambda}}$, by using (iii).

Setting: still considering the case $|\hat{\theta}_i| > \sqrt{2\lambda}$.

Solution: In the solution of (iii) we have proven that

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Exercise 1 (v)

$$\textbf{Objective: } g_i(\theta_i) = -\hat{\theta}_i \theta_i + \frac{\theta_i^2}{2} + \lambda I_{|\theta_i| \neq 0}.$$

(v) Consider the complementary case of (iii) and (iv), i.e., $|\hat{\theta}_i| \leq \sqrt{2\lambda}$, and infer that for the minimizer θ_i^* of g_i it must hold that $\theta_i^* = 0$.

Hint: What is $g_i(0)$? Consider $\tilde{g}_i(\theta) = -\hat{\theta}_i \theta + \frac{\theta^2}{2} + \lambda$, the smooth extension of g_i . What is the relationship between the minimizer of g_i and the minimizer of $\tilde{g}_i$?

(We do not need this hint in the solution presented in the subsequent slides.)

Solution to Exercise 1 (v)

$$\text{Objective: } g_i(\theta_i) = -\hat{\theta}_i \theta_i + \frac{\theta_i^2}{2} + \lambda \mathbf{1}_{|\theta_i| \neq 0}.$$

Same case split as (iii); the comparison flips.

► **Case 1:** $\theta_i = 0$
 $g_i(0) = 0.$

► **Case 2:** $\theta_i \neq 0$, so g_i is a smooth quadratic

$$g_i(\theta_i) = \frac{\theta_i^2}{2} - \hat{\theta}_i \theta_i + \lambda,$$

with minimizer $\theta_i^* = \hat{\theta}_i$ and min value

$$g_i(\hat{\theta}_i) = -\frac{\hat{\theta}_i^2}{2} + \lambda.$$

Compare using $|\hat{\theta}_i|^2 \leq 2\lambda$:

$$\underbrace{-\frac{\hat{\theta}_i^2}{2} + \lambda}_{\text{Case 2 min}} \geq \underbrace{0}_{= g_i(0)}$$

$\Rightarrow$ Case 2 **not smaller** than Case 1; Case 1 wins.

Conclusion: $\theta_i^* = 0$ when $|\hat{\theta}_i| \leq \sqrt{2\lambda}$.

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Conclusion: $\theta_i^* = 0$ when $|\hat{\theta}_i| \leq \sqrt{2\lambda}$.

Solution to Exercise 1 (v)

$$\text{Objective: } g_i(\theta_i) = -\hat{\theta}_i \theta_i + \frac{\theta_i^2}{2} + \lambda I_{|\theta_i| \neq 0}.$$

Same case split as (iii); the comparison flips.

- **Case 1:** $\theta_i = 0$
 $g_i(0) = 0$.
- **Case 2:** $\theta_i \neq 0$, so g_i is a smooth quadratic

$$g_i(\theta_i) = \frac{\theta_i^2}{2} - \hat{\theta}_i \theta_i + \lambda,$$

with minimizer $\theta_i^* = \hat{\theta}_i$ and min value

$$g_i(\hat{\theta}_i) = -\frac{\hat{\theta}_i^2}{2} + \lambda.$$

Compare using $|\hat{\theta}_i|^2 \leq 2\lambda$:

$$\underbrace{-\frac{\hat{\theta}_i^2}{2} + \lambda}_{\text{Case 2 min}} \geq \underbrace{0}_{= g_i(0)}$$

$\Rightarrow$ Case 2 **not smaller** than Case 1; Case 1 wins.

Conclusion: $\theta_i^* = 0$ when $|\hat{\theta}_i| \leq \sqrt{2\lambda}$.

Solution to Exercise 1 (v)

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Conclusion: $\theta_i^* = 0$ when $|\hat{\theta}_i| \leq \sqrt{2\lambda}$.

Putting it all together

From (iii) and (v), per coordinate i :

Strong: $|\hat{\theta}_i| > \sqrt{2\lambda} \implies \theta_i^* = \hat{\theta}_i$ (here $I_{|\hat{\theta}_i| > \sqrt{2\lambda}} = 1$).

Weak: $|\hat{\theta}_i| \leq \sqrt{2\lambda} \implies \theta_i^* = 0$ (here $I_{|\hat{\theta}_i| > \sqrt{2\lambda}} = 0$).

Glue via the indicator (step (iv)):

$$\theta_i^* = \hat{\theta}_i \cdot I_{|\hat{\theta}_i| > \sqrt{2\lambda}}, \quad i = 1, \dots, p.$$

By separability (step (ii)), the joint minimizer assembles coordinate-wise:

$$\hat{\theta}_{L0,i} = \hat{\theta}_i \cdot I_{|\hat{\theta}_i| > \sqrt{2\lambda}}, \quad i = 1, \dots, p$$



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Exercise 2 (a): simulate the data

Task. Build a data set with one informative feature buried in noise.

Draw $n = 100$ observations from

$$Y = \sin(x_1) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

so the response depends on x_1 *only*.

Add $p = 100$ further covariates $x_2, \dots, x_{101}$ that are unrelated to Y .

101 candidate features, but 100 of them are pure noise $\implies$ the regime where regularization earns its keep.

Exercise 2 (a): implementation (Python)

```
1 import numpy as np
2
3 rng = np.random.default_rng(42)
4 n, p_add = 100, 100
5
6 # feature matrix: n rows, (p_add + 1) columns
7 X = rng.standard_normal((n, p_add + 1))
8
9 # only the first column drives Y; the rest are pure noise
10 Y = np.sin(X[:, 0]) + rng.normal(scale=0.5, size=n)
```

► X has shape 100×101 ; only column 0 is signal, columns 1–100 are noise. The Gaussian draw fixes the unspecified ε to $\mathcal{N}(0, 0.5^2)$.

Exercise 2 (b): what to demonstrate

On the simulated data, use models of your choice to illustrate:

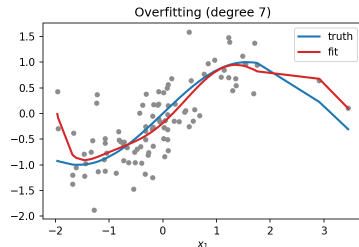
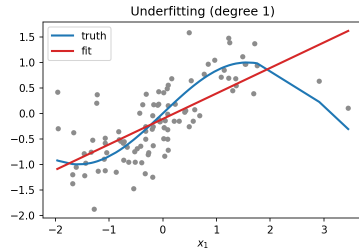
- ▶ under- and over-fitting;
- ▶ $L1$, $L2$ and elastic-net regularization;
- ▶ the underdetermined problem;
- ▶ the bias–variance trade-off;
- ▶ early stopping with a simple neural network.

The R reference uses `glmnet` and `keras`; we port each part to Python with `scikit-learn`.

Exercise 2 (b): under- & over-fitting

```
1 from sklearn.preprocessing import PolynomialFeatures
2 from sklearn.linear_model import LinearRegression
3
4 def poly_fit(x, y, degree, x_eval):
5     pf = PolynomialFeatures(degree, include_bias=False)
6     Xp = pf.fit_transform(x[:, None])
7     m = LinearRegression().fit(Xp, y)
8     return m.predict(pf.transform(x_eval[:, None]))
9
10 yhat1 = poly_fit(x1, Y, 1, xs)    # underfit
11 yhat7 = poly_fit(x1, Y, 7, xs)    # overfit
```

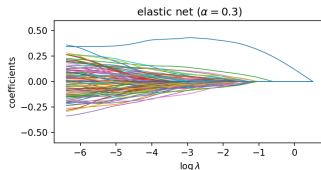
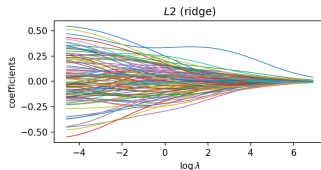
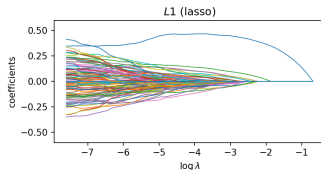
A line is too rigid (high bias); the degree-7 fit starts chasing noise at the edges.



Exercise 2 (b): $L1$, $L2$ & elastic net

```
1 from sklearn.preprocessing import StandardScaler
2 from sklearn.linear_model import (
3     Ridge, lasso_path, enet_path)
4
5 Xs = StandardScaler().fit_transform(X)
6 Yc = Y - Y.mean()
7
8 # L1: lasso coefficient path
9 a1, c1, _ = lasso_path(Xs, Yc)
10 # L2: ridge path over a grid
11 a2 = np.logspace(-2, 3, 100)
12 c2 = [Ridge(alpha=a).fit(Xs, Yc).coef_
13       for a in a2]
14 # elastic net, alpha = 0.3
15 ae, ce, _ = enet_path(Xs, Yc, l1_ratio=0.3)
```

$L1$ sends coefficients to exactly 0 (sparsity); $L2$ shrinks them smoothly; elastic net blends both.



Exercise 2 (b): the underdetermined problem

With $n = 100 < p + 1 = 101$, the Gram matrix $\mathbf{X}^\top \mathbf{X}$ is singular, so the normal equations have no unique solution.

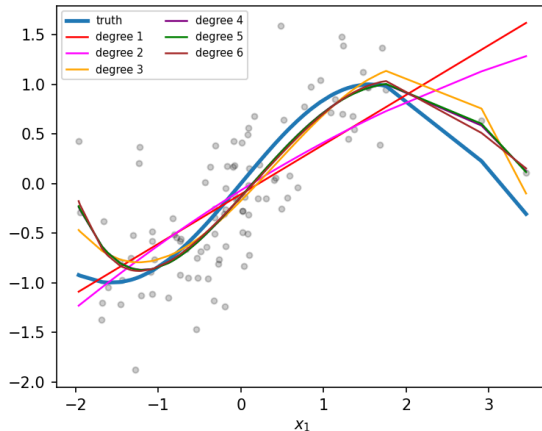
```
1 XtX = X.T @ X                                # shape (101, 101)
2 np.linalg.matrix_rank(XtX)                   # -> 100 (< 101)
3 np.linalg.cond(XtX)                           # -> 2.0e+17
4 beta = np.linalg.solve(XtX, X.T @ Y)         # numerically meaningless
```

R's `solve` aborts with “system is computationally singular”; NumPy returns a vector, but the condition number $\sim 10^{17}$ flags it as garbage.

Exercise 2 (b): bias–variance trade-off

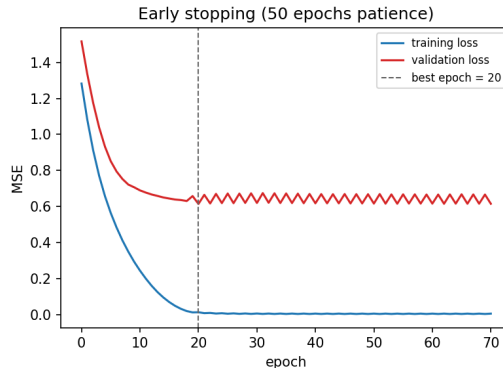
```
1 plt.scatter(x1, Y, alpha=0.2)
2 plt.plot(xs, np.sin(xs), lw=2) # truth
3
4 # overlay fits of growing degree
5 for deg in range(1, 7):
6     yhat = poly_fit(x1, Y, deg, xs)
7     plt.plot(xs, yhat,
8               label=f"degree {deg}")
```

Low degree $\rightarrow$ high bias; high degree $\rightarrow$ high variance (edge wiggles).



Exercise 2 (b): early stopping

```
1 from sklearn.neural_network import MLPRegressor
2 from sklearn.model_selection import
  train_test_split
3 from sklearn.metrics import mean_squared_error
4
5 Xtr, Xv, ytr, yv = train_test_split(
6     X, Y, test_size=0.2, random_state=42)
7 net = MLPRegressor((50, 50), activation="relu",
8     solver="adam", warm_start=True, max_iter
9     =1,
10    random_state=42)
11
12 best, wait, patience = np.inf, 0, 50
13 for epoch in range(300):
14     net.fit(Xtr, ytr)
15     v = mean_squared_error(yv, net.predict(Xv))
16     if v < best - 1e-4:
17         best, wait = v, 0
18     else:
19         wait += 1
20         if wait >= patience:
21             break
```



Training loss keeps dropping while validation loss plateaus $\Rightarrow$ stop at the best epoch.

Summary — connecting the exercise to the lecture

	Key take-away	Lecture slide
Ex 1 (i,ii)	Orthonormal design $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ decouples the regularized ERM into p scalar problems $g_i(\theta_i)$	L1/L2 regularization with orthonormal design
Ex 1 (iii–v)	Each scalar problem $\Rightarrow$ hard threshold $\hat{\theta}_{L0,i} = \hat{\theta}_i \mathbf{1}_{ \hat{\theta}_i > \sqrt{2\lambda}}$	L0 regularization (analog of lasso's soft-thresholding)
Ex 2 (a)	Simulate 1 signal + 100 noise features: the $p \gtrsim n$ regime that needs regularization	Surplus features let the model fit noise instead of signal — the root of overfitting
Ex 2 (b)	Under-/overfitting and the bias–variance trade-off, swept via polynomial degree	Too little capacity misses the signal; too much chases the noise
Ex 2 (b)	L1/L2/elastic-net shrinkage paths; underdetermined OLS; early stopping	L1 zeros out noise features while L2 shrinks smoothly; early stopping $\approx$ implicit L2